# HOT MOVES

Heat is a form of energy that moves from bodies at a higher temperature to those which are colder or at a lower temperature. If two bodies are at the same temperature, then there is no movement of heat form one to the other.
Heat can move from one substance to another in three different ways. What is between the hot and cold substances determines how the heat moves.

Heat can move in three ways

- CONDUCTION
- CONVECTION
- RADIATION

Read pages 52 and 53 of Science Quest 1, before completing the following questions.

**REMEMBER**

1. Complete the table below.

| **Type of heat movement** | **Describe briefly how heat moves** | **In which state or states would the heat move by this method** |
|---|---|---|
| Conduction | | |
| Convection | | |
| Radiation | | |

2. What is an insulator? Name three different materials that can act as insulators.

3. What three things can happen to radiated heat when it arrives at any surface?

**THINK**

1. Think about water as ice, liquid water and water vapour.
   a) In which state of water does convection occur?

   b) In which state of water does conduction occur?

   c) In which state of water would you expect conduction to occur best?

2. Conduction is possible in solids but is not an effective means of heat transfer in liquids or gases. Use the particle model to help you explain why.

3. When you hold a mug of coffee or hot soup, your hands feel warm. How is the heat transferred?

4. When you put your hand on a metal surface and a wooden surface in the same room, one surface feels cooler even though both have the same temperature. Which surface feels cooler? Explain why.

5. If you were to buy a black car and a white car, in which would it be hotter to sit during summer? Why?